2022 秋高数上期末试题（回忆版）

1. Multiple Choice Questions : (only one correct answer for each of the following questions.)

(1) If $f(x)=\frac{ax+b}{x^2-1}$ has a local extreme of 1 at $x=3$, then

(A) $a=3, b=-1$. (B) $a=4, b=-4$. (C) $a=5, b=-7$. (D) $a=6, b=-10$.

(2) Let $f(x)=\frac{\tan x}{|x|(x-\frac{\pi}{2})^4}$. Which of the following statements must be correct?

(A) f is continuous at $x=0$ and f has a jump discontinuity at $x=\frac{\pi}{2}$.

(B) f has a jump discontinuity at $x=0$ and f is continuous at at $x=\frac{\pi}{2}$.

(C) f has an infinite discontinuity at $x=0$ and f has an oscillating discontinuity at $x=\frac{\pi}{2}$.

(D) f has a jump discontinuity at $x=0$ and f has an infinite discontinuity at $x=\frac{\pi}{2}$.

(3) The number of asymptotes of $y=e^{\frac{1}{x^2}}\arctan\frac{x^2+x+1}{(x-1)(x+2)}$ is

(A) 1 (B) 2 (C) 3 (D) 4

(4) Which of the following improper integrals is divergent?

(A) $\int_{-1}^{1}\ln|x|\,dx$ (B) $\int_0^1\frac{e^{\sqrt{x}}}{\sqrt{1-x}}\,dx$ (C) $\int_0^{+\infty}\frac{1}{x^2}\,dx$ (D) $\int_e^{+\infty}\frac{1}{x\ln^2 x}\,dx$.

(5) Suppose that $a<0<b$, and $f(x)$ is continuous on (a,b). Let $F(x)=\begin{cases}\frac{\int_0^x tf(t)\,dt}{x}, & x\neq 0\\ 0, & x=0.\end{cases}$

Which of the following statements must be correct?

(A) F is differentiable on (a,b) and F' is not continuous at $x=0$.

(B) F is differentiable on (a,b) and F' is continuous at $x=0$.

(C) F is not differentiable on (a,b) and F is continuous at $x=0$.

(D) None of the above statements is correct.

2. Fill in the blanks.

(1) The number of the real roots for the equation $x^3-4x^2+x+1=0$ is _____.

(2) If $f(x)=(1+x)(1+2x)\cdots(1+10x)$, then $f'(0)=$ _____.

(3) Use Euler's method to find the approximation for the solution of

$$y'=1+xy,\ y(0)=1$$

Take $dx=0.5$, and start at $x_0=0, y_0=1$. Then $y_2=$ _____.

(4) If $f(x)=\operatorname{arc}\tan\frac{1+x}{1-x}$, then $f'(0)=$ _____.

(5) If $\int_0^1\frac{e^x}{x+1}\,dx=a$, then $\int_0^1\frac{e^x}{(x+1)^2}\,dx=$ _____.

3. Solve the following first-order linear differential equation

$$xy'+2y=x^2+1,\quad x>0.$$

4. (1) Find the area of the region enclosed by the curves $y=x^2-2x$, $y=0$, $x=1$, and $x=3$.

(2) Find the volume of the solid generated by revolving the region in (1) about the y-axis.

5. Assume that f is differentiable at $x=1$, and $f(1)=1, f'(1)=2$. Find the value of $\lim_{n\to\infty}\left(f\left(1+\frac{1}{n}\right)\right)^n$.

6. Evaluate the following limits.

(1) $\lim\limits_{x\to 0^+} \dfrac{\ln \tan 7x}{\ln \tan 2x}$.

(2) $\lim\limits_{n\to\infty} \sum\limits_{k=1}^{n} \dfrac{k}{n^2} \sin^2\left(1+\dfrac{k}{n}\right)$

7. Evaluate the integrals.

(1) $\int x \tan^2 x \, dx$

(2) $\int_0^{+\infty} \dfrac{\tan^{-1} e^x}{e^{2x}} \, dx$.

(3) $\int_1^{\sqrt{3}} \dfrac{1}{x(1+x^2)^2} \, dx$

(4) $\int \dfrac{\ln(1-x^2)}{x^2\sqrt{1-x^2}} \, dx$.

8. Assume $f(x)$ is continuous on $[0,1]$ and differentiable on $(0,1)$. If $f(0)=f(1)=0$, $f\left(\frac{1}{2}\right)=1$, prove that:

(1) there exists $c\in\left(\frac{1}{2},1\right)$, such that $f(c)=c$.

(2) For any real number k, there always exists $\xi\in(0,c)$, such that $f'(\xi)-k|f(\xi)-\xi|=1$.